

Møller Polarimetry Update

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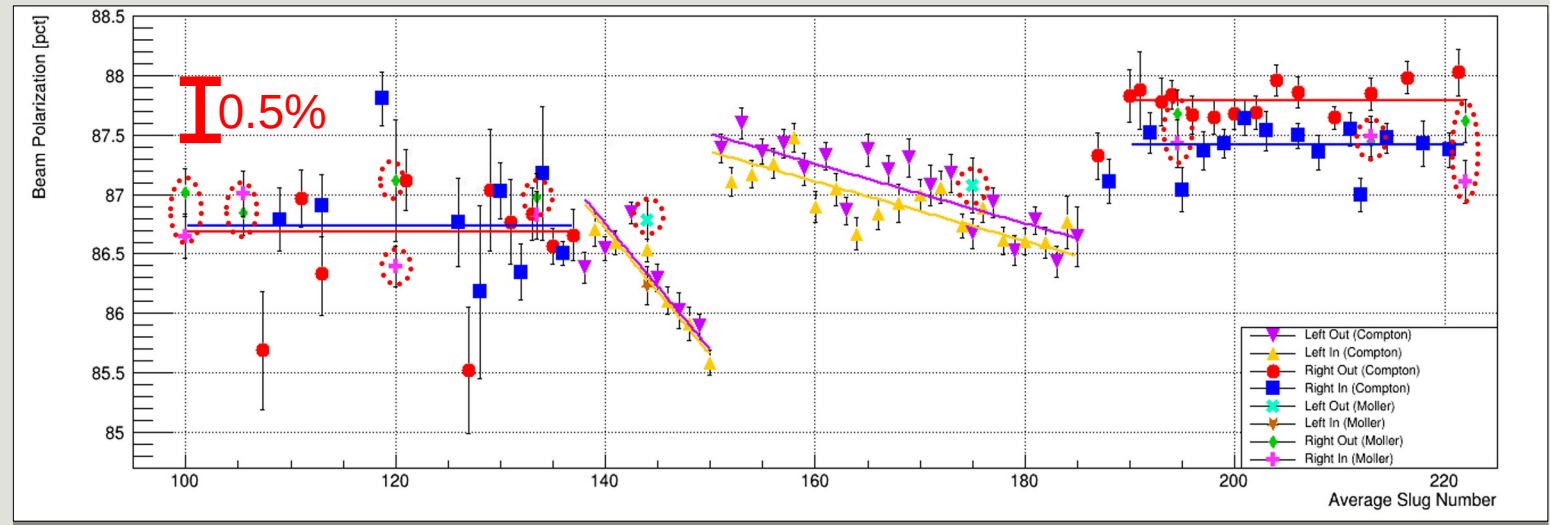
December 2021 :: MOLLER Collaboration
Meeting



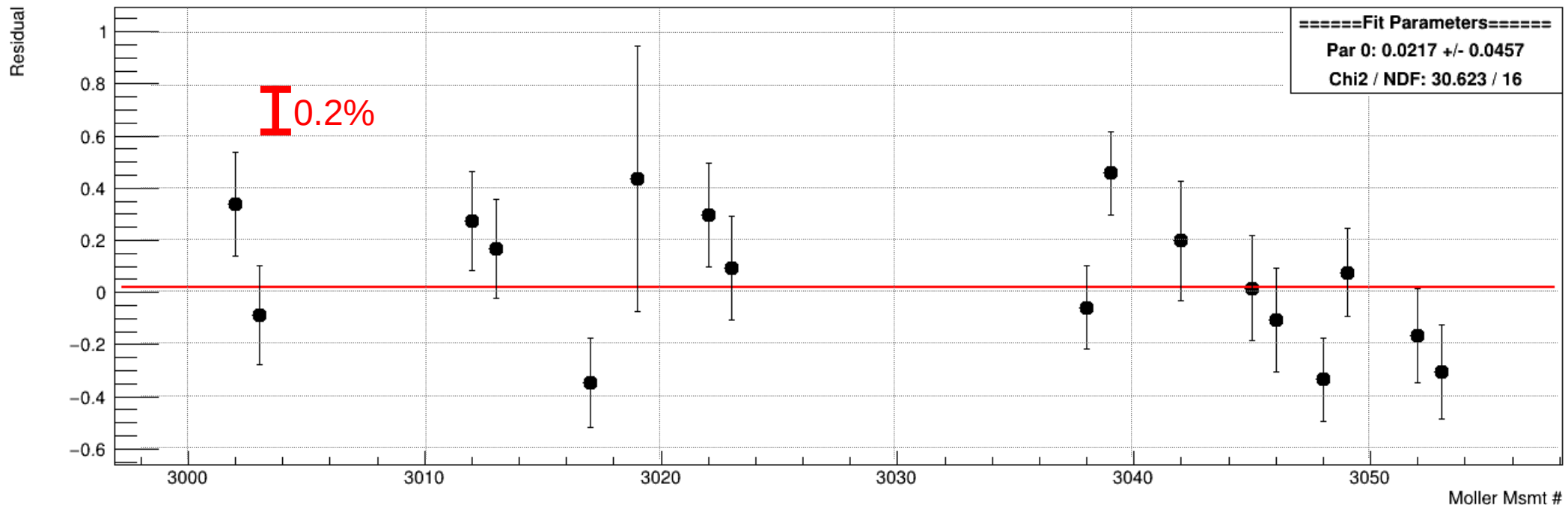
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CREX Results

Møller and Compton
measurements
during CREX
Agreement with Compton
at $\pm 0.02\%$ level



Moller Residuals with Compton Fits

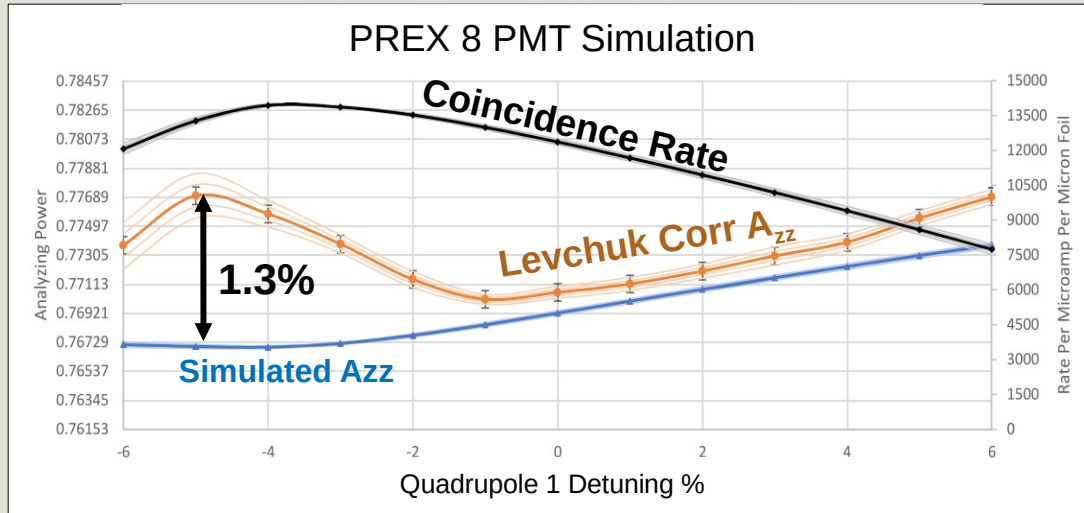


CREX Systematics Results

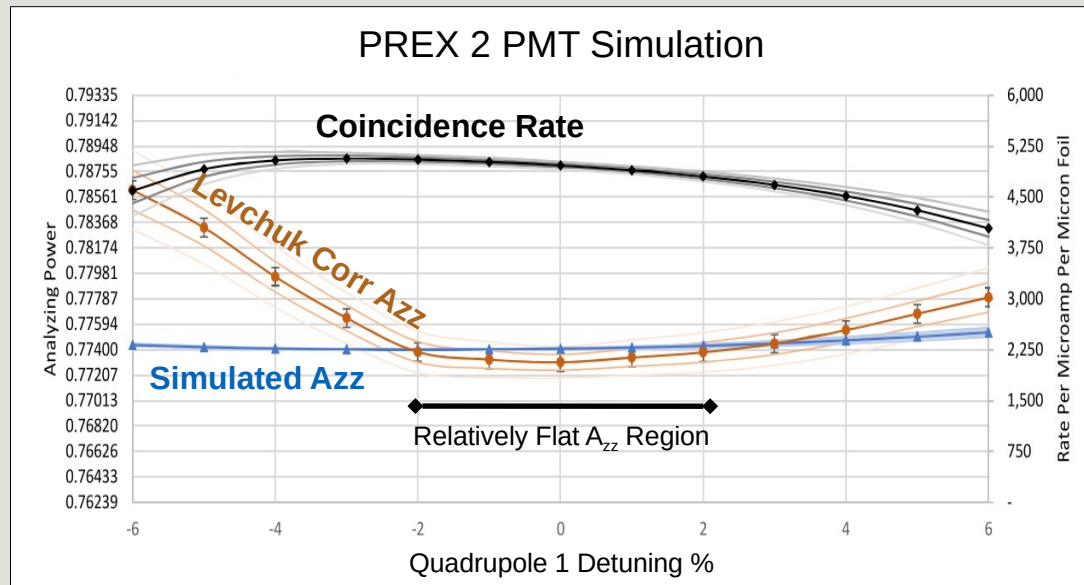
- Record accuracy for JLab Møller polarimetry.
- Significant progress has been made on A_{zz} systematic.
- It looks like we will be able to reach 0.14% or better uncertainty for A_{zz} at 11 GeV.

Source	Value	$\delta P/P$ (%)	Goal (%)
A_{zz}	0.75421	0.16	0.14
Foil Polarization	0.08005	0.57	0.30
Dead Time Correction	0.148%	0.15	0.05
Accidental Correction	0.205%	0.04	0.04
Charge Normalization	0.029%	0.01	0.01
Null Asymmetry (Cu Foil)	0.0%	0.22	0.05
PITA Variation	—	0.06	0.06
Spin Precession (dP/P)	—	0.04	0.10
High Current Extrapolation	—	0.50	0.20
Bleed through	—	0.18	0.00
Slit Dependence	—	0.10	0.00
Total		0.85	0.41

Moller A_{zz} : Two Recent Improvements



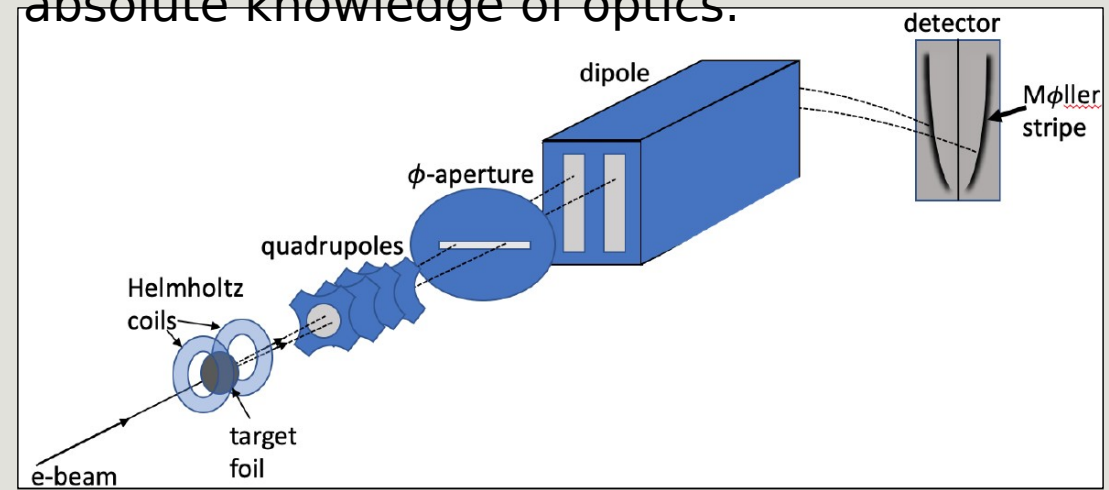
Active
PMTs
(Red)



Active
PMTs
(Red)

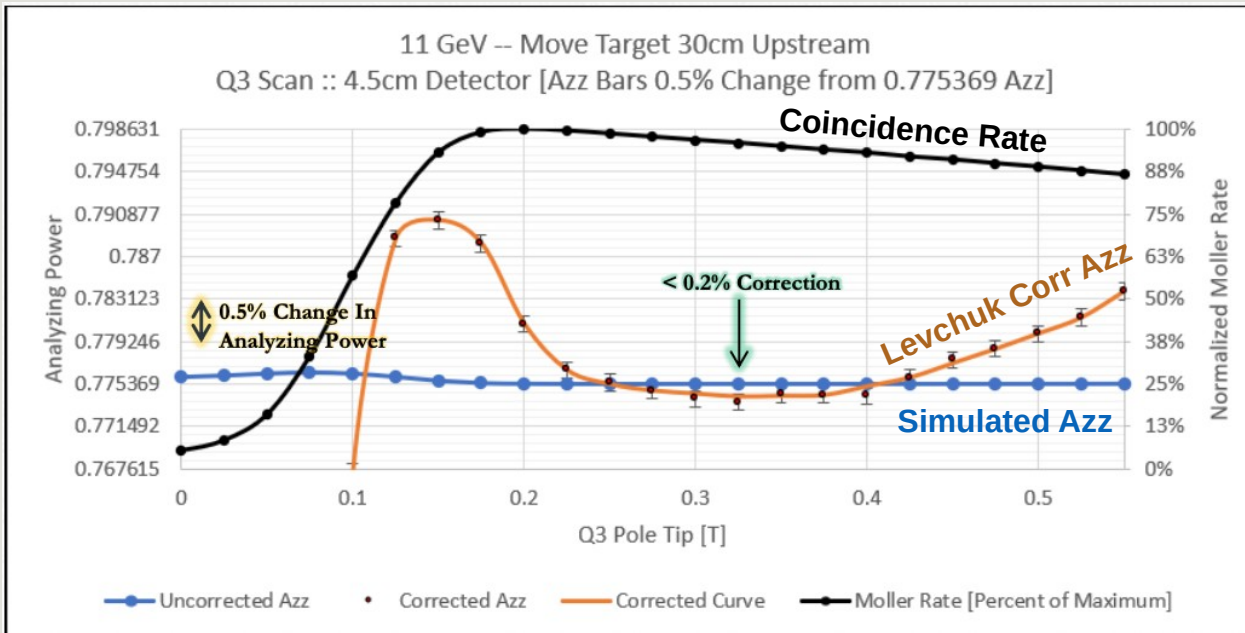
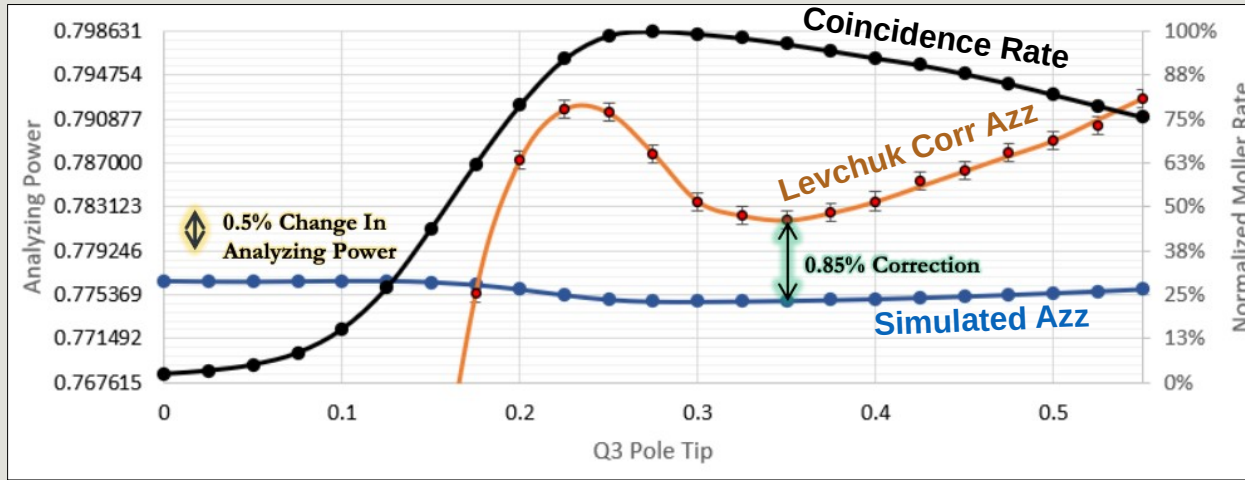


Our knowledge of spectrometer optics (field maps) is limited to several percent uncertainty, BUT simulation shows if we calibrate our settings relative to the rate maximum, we are highly insensitive to absolute knowledge of optics.



Limiting the vertical detector acceptance allows AND over-tuning past rate peak to pull electrons off dipole walls allowed us to tune the Levchuk correction to nearly zero AND create a large flat region of A_{zz} .

A_{zz} at 11 GeV optics



At 11 GeV, with current beamline setup, we are unable to tune away large Levchuk correction nor create a flat region insensitive to accurate knowledge of the optical tune. However, moving target 30 cm upstream accomplishes what we want by allowing Moller scatters to further separate before quadrupole steering.

Caveat: results for 4.5cm vertical detector acceptance. May require detector redesign or collimator.

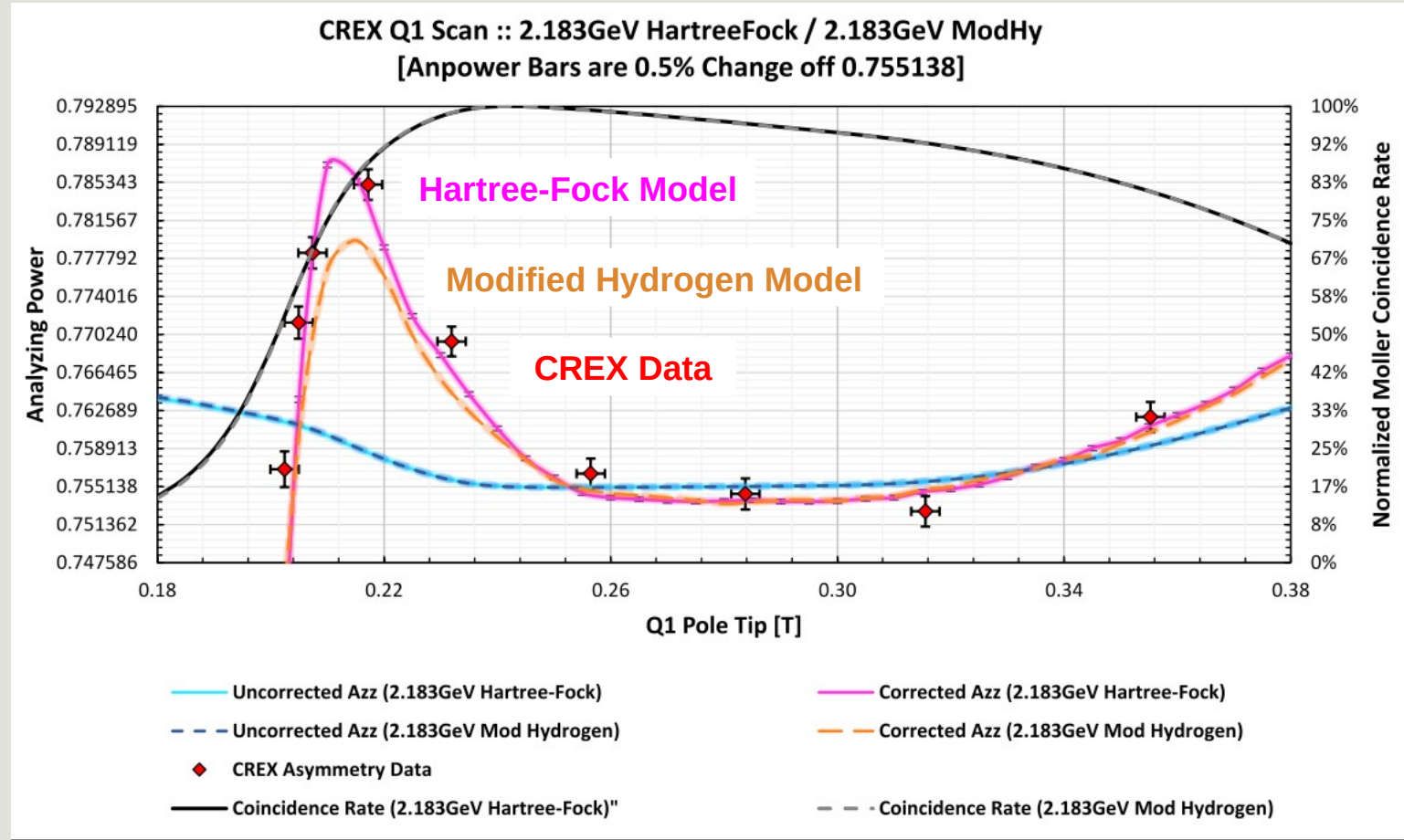
Now officially

Beam Line Redesign Being Implemented

- Moving Møller target coils upstream required some re-configuring of the beam line design: design changes now included in model
- Plan in place to implement improved MOLLER beam line including Møller target move before MOLLER
 - Could be as early as Spring 2022, although the probability for getting all parts in hand on time is low.
 - Early implementation gives us time to study impact of redesign and benchmark simulation ahead of time.

Another recent improvement for A_{zz}

- Levchuk correction requires a model of the momentum distributions for polarized and unpolarized electrons in Fe
- For past two decades this correction has been done using distributions from modified hydrogen wavefunctions.
- Data from CREX showed that this calculation predicted a 30% smaller “peak” than measured.
- Reached out to expert John Perdew* at Temple University and his student Aaron Kaplan suggested a Hartree-Fock model and provided the momentum distributions.



John Perdew is one of the most highly cited scientists of all time with over 150,000 Google citations

CREX Troublesome Systematics

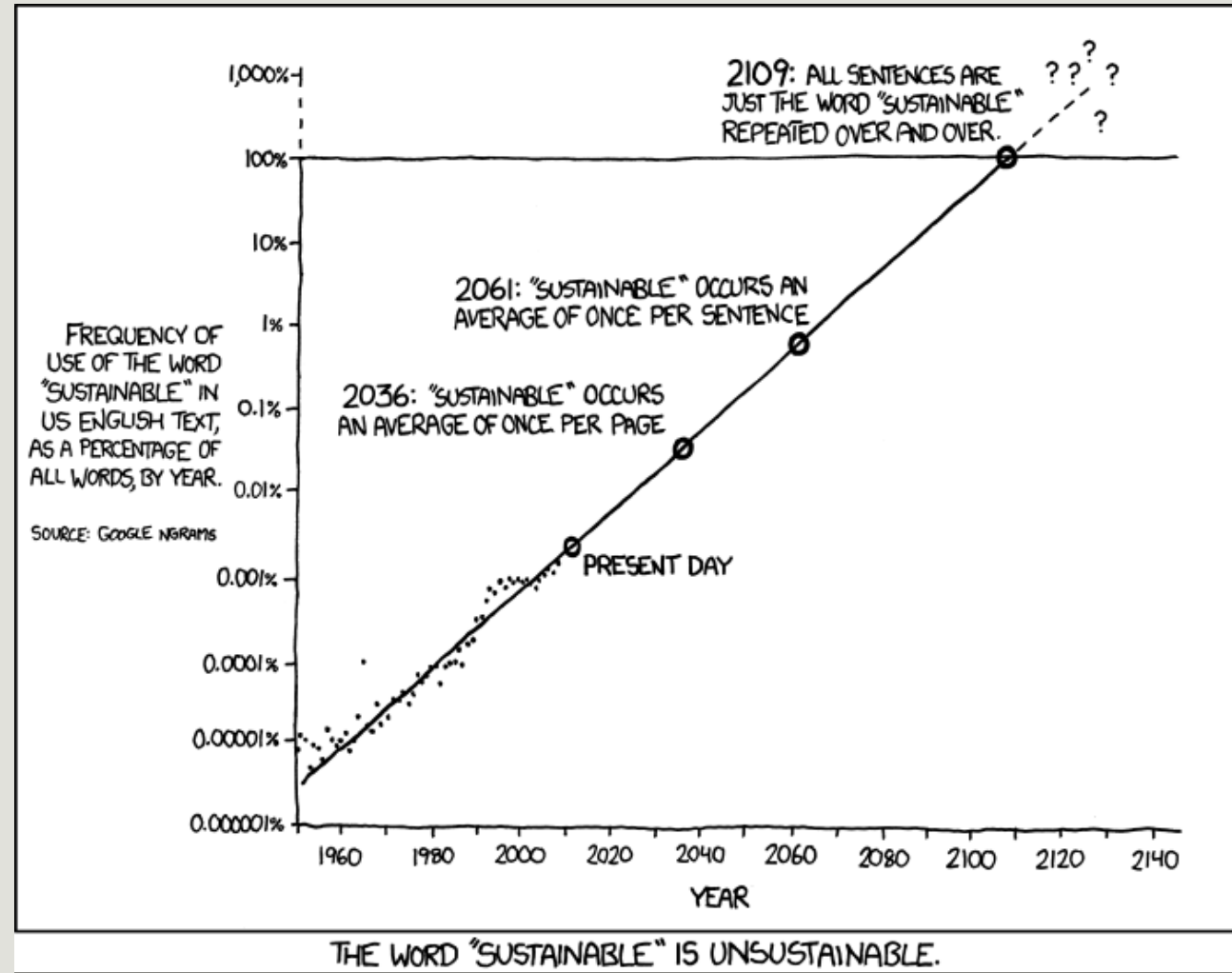
Systematics inflated by foil polarization and high current extrapolation

- These two contribute a total of 0.76% and the rest is 0.38%
- Likely overestimated (given level of agreement with Compton which had 0.5% uncertainty) but difficult to limit without better measurements

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Extrapolation to High Current

- Potential to be the limiting systematic unless we measure it.
- We will develop a plan to directly measure:
 - Beat frequency method used before where the laser is tuned just off the 499 MHz so only every nth pulse actually goes to the hall but full power on photocathode.
 - Can investigate what statistics Compton could get at 5 μA for

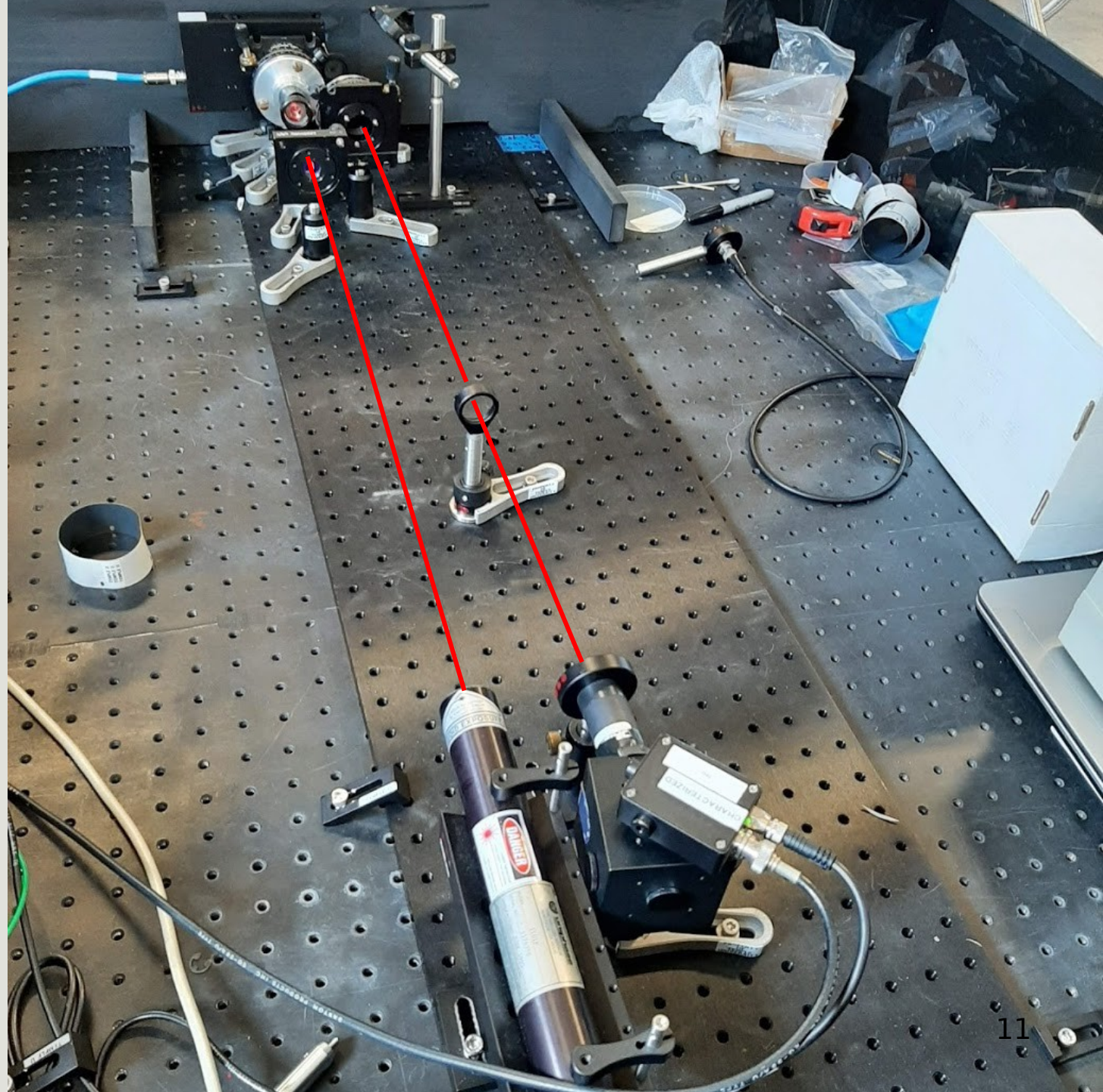
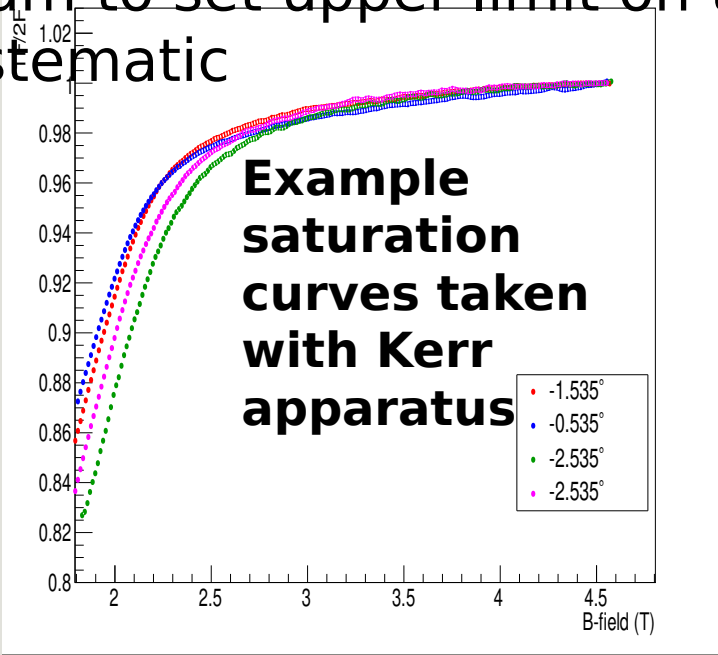


Foil Polarization

- Publication nearly ready to submit to NIM that shows Fe foil polarization at saturation known to $\pm 0.24\%$
- Still need to prove that we are at saturation
- Sensitivity of degree of saturation on foil and angle and flatness needs to be measured
- Potentially large systematic: 1% smaller polarization during PREX-2 when running on a wrinkle
- Attempts to measure with Kerr apparatus only partially successful.
- Kerr apparatus: modulate polarization at fixed frequency and pull out polarization signal on reflected light from magnetized foil with lock-in amplifier (magnetization rotates polarization)

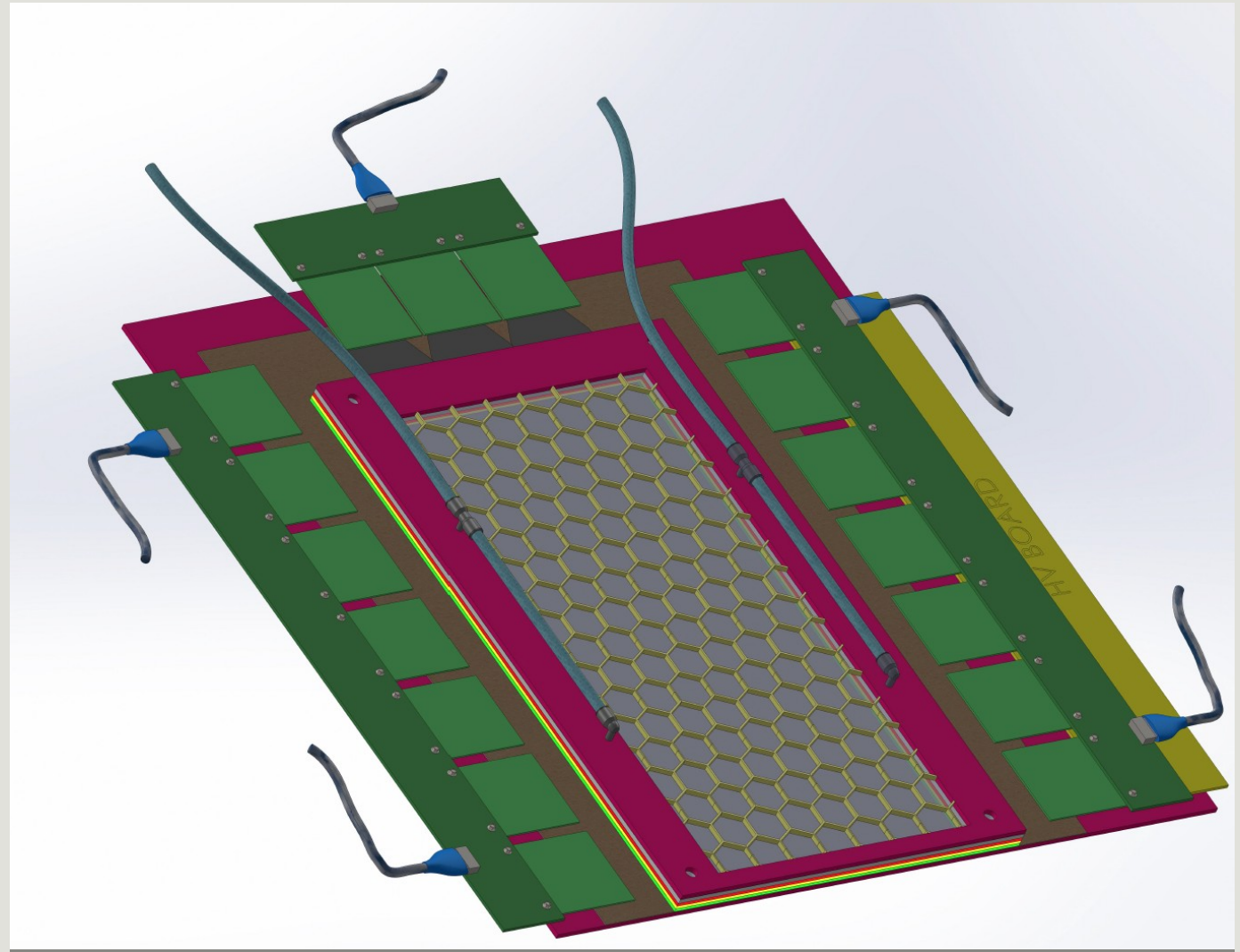
Kerr Apparatus

- Not able to conclusively reach saturation but see signature of interference effects.
- Longer careful study required but may not be possible.
- Need to develop plan for using beam to set upper limit on this systematic



GEMS for the Møller Polarimeter

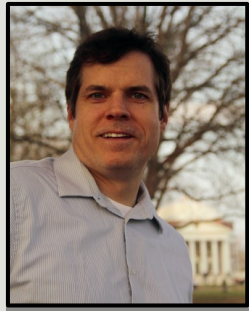
- GEMs are being installed between the dipole and the detector to help verify simulated radiative, Levchuk and multiple scattering effects.
- Design recently completed by Nilanga's group at UVA -- ready for construction
- Readout split into right and left required to allow for resolution of left and right coincident Moller electrons



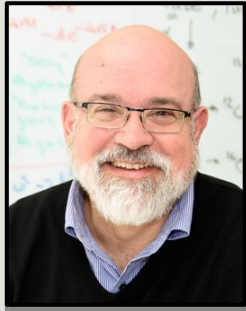
Collimator

- Addition of a collimator is part of the project to help limit systematics
- Need more simulation work to decide if this is needed and what its design should be.
 - Initial investigation underway
- May be required to limit vertical acceptance on detector (remove sensitivity to Levchuk and knowledge of quad optics settings)
- Target date set to make design decision by early April

Møller Polarimetry Active Personnel



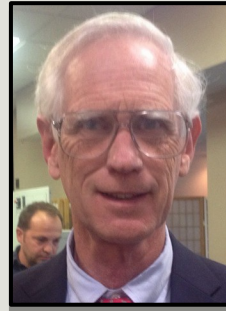
**Donald
Jones**



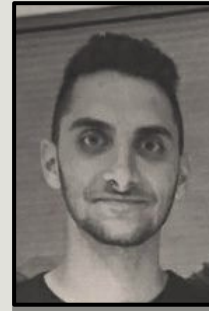
**Jim
Napolitano**



**Eric
King**



**Paul
Souder**



**Faraz
Chahili**



**David
Gaske
II**



**Simona
Malace**



**Bill
Henry**



**Kent
Paschke**

- Working on polarized materials review.
- Continued systematics reduction.
- Reviewing design issues.

- Working on GEM insights.
- GEM event reconstruction of simulated

- Maintain hardware.
- Coordinate project deadlines.
- Technical operators.

- Insight
- Advice

Summary

- Encouraging progress made on systematics during PREX-2 and CREX
- Progress on A_{zz} & Levchuk systematics nearly meet MOLLER requirements
- Two systematics (foil polarization and high current extrapolation) are highest priorities and will likely require beam time to limit
- Progress is being made on hardware improvements included in the MOLLER project, with
 - GEM design complete and support structure design to begin soon
 - Collimator design choice made by early Spring.
- All systematics being reviewed carefully